1. Aim of the practical: To understand and demonstrate managing SE Linux Security

2. Objective: The goal is to effectively understand and manage SELinux by checking its current status, adjusting its operational mode (Enforcing, Permissive, or Disabled) in the configuration file, and applying these changes through a system reboot, ensuring the correct security policies are enforced according to the selected mode.To manage file permissions and control access to files for specific users and groups to ensure data security.

3. Hardware and Software Requirements:

Computer System: Host system running virtualization software.

Operating System: OS on the virtual machine (e.g., Fedora).

4. Commands/Steps:

Step-1: : Check SELinux status: Used to verify current SELinux mode.

```
asr@fedora:~$ sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33
asr@fedora:~$
```

Step-2: Open config file: Access SELinux settings file.

```
asr@fedora:~$ sudo vim /etc/selinux/config
[sudo] password for asr:
```

Step-3: Locate SELINUX= line: Used to find mode setting to change.

```
#
# NOTE: In earlier Fedora kernel builds, SELINUX=disabled would also
# fully disable SELinux during boot. If you need a system with SELinux
# fully disabled instead of SELinux running with no policy loaded, you
# need to pass selinux=0 to the kernel command line. You can use grubby
# to persistently set the bootloader to boot with selinux=0:
#
#    grubby --update-kernel ALL --args selinux=0
#
# To revert back to SELinux enabled:
#
#    grubby --update-kernel ALL --remove-args selinux
#
SELINUX=enforcing
# SELINUXTYPE= can take one of these three values:
#     targeted - Targeted processes are protected,
#     minimum - Modification of targeted policy. Only selected processes are pro
tected.
#     mls - Multi Level Security protection.
SELINUXTYPE=targeted
```

```
#
# NOTE: In earlier Fedora kernel builds, SELINUX=disabled would also
# fully disable SELinux during boot. If you need a system with SELinux
# fully disabled instead of SELinux running with no policy loaded, you
# need to pass selinux=0 to the kernel command line. You can use grubby
# to persistently set the bootloader to boot with selinux=0:
#
#    grubby --update-kernel ALL --args selinux=0
#
# To revert back to SELinux enabled:
#
#    grubby --update-kernel ALL --remove-args selinux
#
SELINUX=permissive
# SELINUXTYPE= can take one of these three values:
#     targeted - Targeted processes are protected,
#     minimum - Modification of targeted policy. Only selected processes are pro
tected.
#     mls - Multi Level Security protection.
SELINUXTYPE=targeted


~
-- INSERT --                                          22,19         Bot
```

Step-4: Save and exit: Used to Store changes in the config.

```
#
# NOTE: In earlier Fedora kernel builds, SELINUX=disabled would also
# fully disable SELinux during boot. If you need a system with SELinux
# fully disabled instead of SELinux running with no policy loaded, you
# need to pass selinux=0 to the kernel command line. You can use grubby
# to persistently set the bootloader to boot with selinux=0:
#
#    grubby --update-kernel ALL --args selinux=0
#
# To revert back to SELinux enabled:
#
#    grubby --update-kernel ALL --remove-args selinux
#
SELINUX=permissive
# SELINUXTYPE= can take one of these three values:
#     targeted - Targeted processes are protected,
#     minimum - Modification of targeted policy. Only selected processes are pro
tected.
#     mls - Multi Level Security protection.
SELINUXTYPE=targeted


~
:wq!
```

Step-5: Verify current SELinux mode.

```
asr@fedora:~$ sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33
asr@fedora:~$
```

5. Learning outcomes (What I have learned):

1. SELinux operates in three modes: Enforcing (actively applies security policies), Permissive (logs policy violations without enforcing them), and Disabled (completely disables SELinux).
2. SELinux configuration is controlled via the /etc/selinux/config file, where you can adjust the mode by modifying the SELINUX= line.
3. Any changes to the SELinux mode will take effect after a system reboot.

Evaluation Grid (To be created as per the SOP and Assessment guidelines by the faculty):

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet		8 Marks
2.	Viva		10 Marks
3.	Simulation		12 Marks
	Total		30 Marks

Teacher Signature